As the number of different ligands increases, the number of possible isomers also increases. For the octahedral complex of the type $[MABCDEF]^{n\pm}$, where A, B, C, D, E and F are monodentate ligands, fifteen different orientation are possible corresponding to 15 geometrical isomers. It is difficult to generate all the possible isomers.

Evaluate yourself 5:

a. Three compounds A ,B and C have empirical formula $CrCl_3.6H_2O$. they are kept in a container with a dehydrating agent and they lost water and attaining constant weight as shown below. Identify the molecular formula of compounds.

Compound	Initial weight of the compound(in g)	Constant weight after dehydration (in g)
A	4	3.46
B	0.5	0.466
C	3	3

b. Indicate the possible type of isomerism for the following complexes and draw their isomers

(i) $[Co(en)_3][Cr(CN)_6]$ (ii) $[Co(NH_3)_5(NO_2)]^{2+}$ (iii) $[Pt(NH_3)_3(NO_2)]Cl$

5.5.3 Optical Isomerism

Coordination compounds which possess chairality exhibit optical isomerism similar to organic compounds. The pair of two optically active isomers which are mirror images of each other are called enantiomers. Their solutions rotate the plane of the plane polarised light either clockwise or anticlockwise and the corresponding isomers are called 'd' (dextro rotatory) and 'l' (levo rotatory) forms respectively. The octahedral complexes of type $[M(xx)_3]^{n\pm}$, $[M(xx)_2AB]^{n\pm}$ and $[M(xx)_2B_2]^{n\pm}$ exhibit optical isomerism.

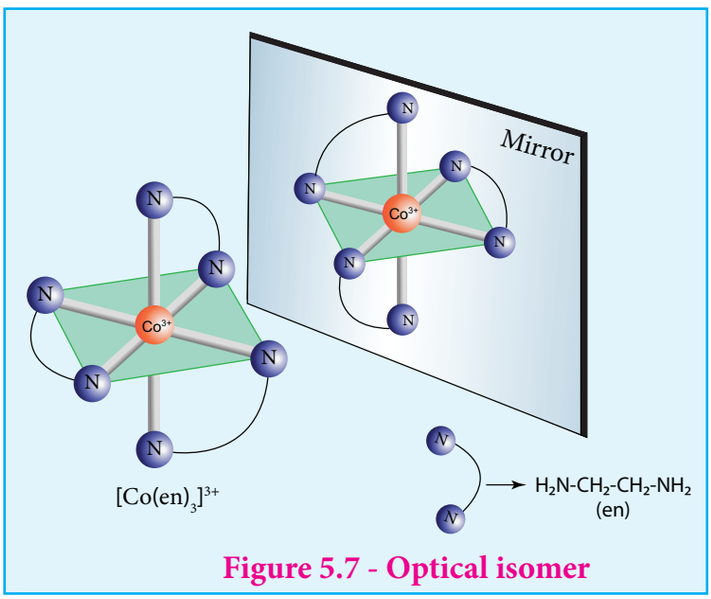


Figure 5.7 - Optical isomer

Examples:

The optical isomers of $[Co(en)_3]^{3+}$ are shown in figure 5.7.

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